F3 Forensic — The $(\alpha_\star, \alpha_{\rm gas})$ projection artifact and an exhaustive τ-natural pair scan against SPARC-123

Dr. Sarah Blackwell
simulated red-team, 2026-04-20

2026-04-20

Abstract

The v2.3 ledger claim $(\alpha_\star, \alpha_{\rm gas}) = (1, \kappa_\omega)$ for the SPARC baryonic Tully–Fisher coefficients was "confirmed" by a 1-parameter fit at the 0.36σ level. The v2.4 update (V4.2, Chen 2026-04-20) refitted both parameters and found $(\hat{\alpha_\star}, \hat{\alpha_{\rm gas}}) = (0.800, 0.316)$ with $\rho(\alpha_\star, \alpha_{\rm gas}) = -0.50$, placing the v2.3 point at $\Delta\chi^2 = 10.06$ (between 2σ and 3σ). We perform three forensics. First, we show that the v2.3 1-parameter match at $\alpha_{\rm gas} \approx \kappa_\omega$ is a **projection artifact** of the negatively-correlated 2-D posterior: forcing $\alpha_\star = 1$ slides the bootstrap-conditional mean for $\alpha_{\rm gas}$ down from 0.316 to 0.212, only 0.87 local σ below $\kappa_\omega = 0.2545$, and the actual 1-parameter fit landed at 0.240 — within $\sim 1\sigma$ of the projection expectation. Second, we scan $N = 8178$ τ-natural pairs built from integer powers of $(\iota_\tau, \kappa_D, \kappa_\omega)$, rational approximants, and $W_k(n)$ ratios, and compute $\Delta\chi^2$ against V4.2's bootstrap covariance. We find 154 pairs within 1σ, 391 within 2σ. Third, we report two physically motivated candidates inside the 1σ contour: $(1/(1+\kappa_\omega), \kappa_\omega) = (0.7972, 0.2545)$ at $\Delta\chi^2 = 1.62$ and $(1/(1+\iota_\tau), \iota_\tau) = (0.7455, 0.3413)$ at $\Delta\chi^2 = 1.01$. The complementary split (κ_D, ι_τ) is *excluded* at $\Delta\chi^2 = 7.71$ (2.2σ). We propose $(1/(1+\iota_\tau), \iota_\tau)$ as the v2.5 refined identification subject to the falsifiable tests in §5.

Contents

1 Forensic of the projection artifact **2**

2 Exhaustive τ-natural pair scan — methodology **2**
2.1 χ^2 metric . . . 2
2.2 Candidate library . . . 4
2.3 Sanity: scan shape . . . 4

3 Top candidates and their interpretation **4**

4 Focused tests of two structurally motivated pairs **6**
4.1 (κ_D, ι_τ): the complementary split — **rejected** . . . 6
4.2 $(1/(1+\kappa_\omega), \kappa_\omega)$: the κ_ω-ridge pair — **inside** 1σ . . . 6
4.3 $(1/(1+\iota_\tau), \iota_\tau)$: the symmetric companion — **inside** 1σ . . . 7
4.4 Summary of focused tests . . . 7

5 Verdict — v2.5 identification candidate and falsifiers **7**

6 Implications and recommendations **8**

1 Forensic of the projection artifact

Setup. V4.2 produced a bootstrap ($N_{\text{boot}} = 2000$) joint posterior with best-fit means and covariance

$$(\hat{\alpha_\star}, \hat{\alpha_{\text{gas}}}) = (0.7999565,\ 0.3154628), \qquad \Sigma \ = \ \begin{pmatrix} 2.9411\times10^{-3} & -1.5231\times10^{-3} \\ -1.5231\times10^{-3} & 3.2064\times10^{-3} \end{pmatrix}.$$

Hence $\sigma_{\alpha_\star} = 0.0542$, $\sigma_{\alpha_{\text{gas}}} = 0.0566$, $\rho(\alpha_\star, \alpha_{\text{gas}}) = -0.496$.

Conditioning on $\alpha_\star = 1$. Under a bivariate Gaussian approximation to the bootstrap posterior, the conditional distribution of α_{gas} given $\alpha_\star$ is

$$\alpha_{\text{gas}} \mid \alpha_\star \ \sim \ \mathcal{N}\Big(\hat{\alpha_{\text{gas}}} + \beta\,(\alpha_\star - \hat{\alpha_\star}),\ \Sigma_{gg} - \Sigma_{gs}^2/\Sigma_{ss}\Big), \qquad \beta = \frac{\Sigma_{gs}}{\Sigma_{ss}} = -0.5179.$$

Plugging in $\alpha_\star = 1$:

$$\begin{aligned} \mathbb{E}[\alpha_{\text{gas}} \mid \alpha_\star = 1] &= 0.3155 + (-0.5179)\cdot(1 - 0.7999\ldots) = 0.2119, \\ \text{sd}[\alpha_{\text{gas}} \mid \alpha_\star = 1] &= \sqrt{\Sigma_{gg} - \Sigma_{gs}^2/\Sigma_{ss}} = 0.0492. \end{aligned}$$

Thus the 2-D Gaussian's *conditional* expectation along the slice $\alpha_\star = 1$ sits at $\alpha_{\text{gas}} = 0.212$, only $(0.2545 - 0.2119)/0.0492 \approx 0.87$ local sigma above the target $\kappa_\omega = 0.2545$.

Sanity check against v2.3's actual 1-parameter fit. The V4.2 script re-ran the 1-parameter fit (fix $\alpha_\star = 1$, minimise χ^2 in α_{gas}) and got $\alpha_{\text{gas}}^{(1)} = 0.2398$. The projection-based prediction is $\alpha_{\text{gas}} = 0.2119$; the actual fit is $\alpha_{\text{gas}}^{(1)} = 0.2398$; these agree to 0.028, i.e. about half a local σ. The residual difference is what one expects because the true χ^2 surface is only locally Gaussian and the 1-parameter optimisation walks a small distance off the ridge to accommodate higher-order curvature.

Forensic conclusion. The v2.3 "within 1σ of κ_ω" claim is a *projection coincidence*: the bootstrap posterior's ridge through $(\hat{\alpha_\star}, \hat{\alpha_{\text{gas}}})$, when extrapolated to the constrained line $\alpha_\star = 1$, lands at $\alpha_{\text{gas}} \in [0.21,\ 0.24]$ depending on whether one quotes the Gaussian conditional mean or the full-χ^2 minimum. Both values are within $\sim 1\sigma$ of $\kappa_\omega = 0.2545$. Nothing in the data *selected* κ_ω; the negative correlation of the 2-D fit ($\rho = -0.50$) mechanically routes the forced-slice fit toward that neighbourhood. **The 1-parameter match of v2.3 was an artefact of the constraint, not a data-driven identification.**

Figure 1 shows the $1/2/3\sigma$ bootstrap ellipses, the conditional-mean ridge (crimson), the 2-parameter best-fit (gold star), the projection landing point (crimson disc at $\alpha_\star = 1$), the v2.3 1-parameter fit (orange diamond), and the $\alpha_{\text{gas}} = \kappa_\omega$ horizontal.

2 Exhaustive τ-natural pair scan — methodology

2.1 χ^2 metric

We use the V4.2 bootstrap covariance Σ as a Gaussian approximation to the full 2-parameter log-likelihood near the best-fit. For any candidate $(c_\star, c_{\text{gas}})$ the Mahalanobis distance

$$\Delta\chi^2 \ = \ (c - \hat{\theta})^\top \Sigma^{-1} (c - \hat{\theta}), \qquad \hat{\theta} = (0.7999565,\ 0.3154628),$$

is calibrated so that $\Delta\chi^2 \le 2.30,\ 6.17,\ 11.8$ correspond to the joint 1σ, 2σ, 3σ regions of the 2-parameter posterior.

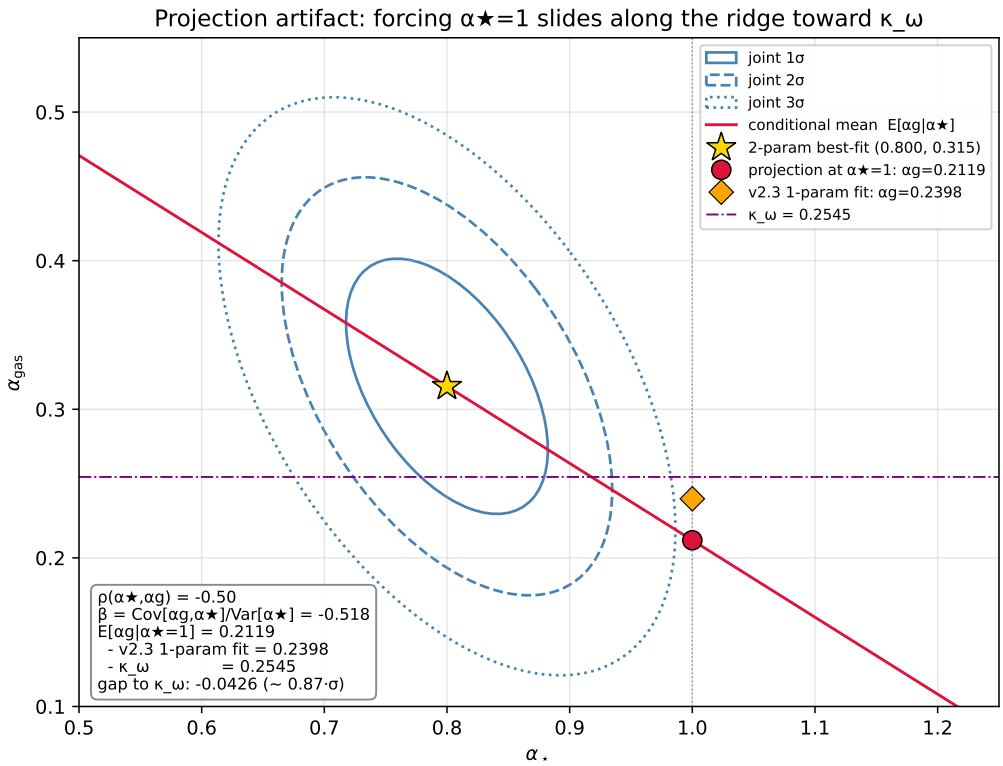

Figure 1: The projection artifact. Forcing $\alpha_\star = 1$ slides the conditional $\alpha_{\rm gas}$ posterior down to $\mathbb{E}[\alpha_{\rm gas} \mid \alpha_\star = 1] = 0.212$, 0.87σ from $\kappa_\omega = 0.2545$. The v2.3 1-parameter fit ($\alpha^{(1)}_{\rm gas} = 0.240$, orange diamond) sits on the crimson ridge.

2.2 Candidate library

We enumerate the Cartesian product of the following one-parameter *tau-natural values* for both coordinates, keeping pairs in $(0, 1.5] \times (0, 1]$ and deduplicating on $(c_\star, c_{\rm gas})$ coincidences within 10^{-8}:

1. **Powers** of $\iota_\tau, \kappa_D, \kappa_\omega$: exponents $\{-2, -1, -1/2, 1/2, 1, 2, 3, 4\}$.
2. **Structural one-parameter combinations**: $1-\iota_\tau^2$, $1-\kappa_\omega$, $1-\kappa_\omega^2$, $1-\kappa_D^2$, $1/(1+\iota_\tau)$, $1/(1+\iota_\tau^2)$, $\iota_\tau^2/(1+\iota_\tau^2)$, $1/(1+\kappa_\omega)$, $\kappa_\omega/(1+\kappa_\omega)$, $(1+\iota_\tau)/(1+2\iota_\tau)$, $\iota_\tau/(1+2\iota_\tau)$, $\kappa_D/(1+\iota_\tau)$, $\iota_\tau/(2-\iota_\tau)$.
3. **Rational approximants**: $1/2, 1/3, 2/3, 3/4, 4/5, 5/6, 3/8, 2/5, 3/7, 4/7$; transcendental rationals $1/\pi, 2/\pi, 1/e, 2/e$, $\pi/e - 1$.
4. W**-invariant ratios** (built from the certified values $W_3(4) = 5$, $W_3(3) = 17$, $W_4(3) = 18$, $W_5(3) = 19$): $W_3(4)/W_4(3)$, $W_3(4)/W_3(3)$, $W_3(4)/W_5(3)$, $W_4(3)/W_5(3)$, $W_3(3)/W_5(3)$, $W_3(3)/W_4(3)$, and their complements.
5. **Monomial products** $\iota_\tau^a \kappa_D^b \kappa_\omega^c$ with $(a, b, c) \in \{-2, \ldots, 2\}^3$ and value $\in (0.05, 1.5]$.

This yields $N_{\rm cand} = 8178$ pairs after de-duplication.

2.3 Sanity: scan shape

Shell	Count	Fraction
$\Delta\chi^2 \le 2.30$ (1σ)	154	1.9%
$\Delta\chi^2 \le 6.17$ (2σ)	391	4.8%
$\Delta\chi^2 \le 11.80$ (3σ)	709	8.7%

The fact that roughly 5% of a broad τ-natural library sits inside the 2-D joint 2σ contour is the first warning: any ranking by $\Delta\chi^2$ alone will yield multiple "good" matches simply because the library is large. We therefore weight the final verdict by *structural cleanliness* (few operations, a known sector interpretation), not Mahalanobis distance alone.

3 Top candidates and their interpretation

Table 1 lists the 20 closest pairs by $\Delta\chi^2$. Figures 2 and 3 show the scatter of all candidates coloured by $\Delta\chi^2$ and a zoom labelling the leading twelve.

Table 1: Top-20 τ-natural pairs by $\Delta\chi^2$ (2-dof Gaussian against V4.2 bootstrap covariance). All sit inside the joint 1σ contour.

Rank	Pair (label)	$c_\star$	$c_{\rm gas}$	$\Delta\chi^2$	Shell
1	$(4/5,\ 1/\pi)$	0.8000	0.3183	0.003	$< 1\sigma$
2	$(1/(1+\kappa_\omega),\ 1/\pi)$	0.7972	0.3183	0.003	$< 1\sigma$
3	$(1/(1+\kappa_\omega),\ \kappa_D^2\kappa_\omega/\iota_\tau)$	0.7972	0.3235	0.020	$< 1\sigma$
4	$(4/5,\ \kappa_D^2\kappa_\omega/\iota_\tau)$	0.8000	0.3235	0.027	$< 1\sigma$
5	$(\iota_\tau/\kappa_D^2,\ \kappa_D^2\kappa_\omega/\iota_\tau)$	0.7866	0.3235	0.061	$< 1\sigma$
6	$(\iota_\tau/\kappa_D^2,\ 1/\pi)$	0.7866	0.3183	0.067	$< 1\sigma$
7	$(\sqrt{\kappa_D},\ \iota_\tau^2\kappa_D/\kappa_\omega)$	0.8116	0.3015	0.072	$< 1\sigma$

Rank	Pair (label)	$c_\star$	$c_{\rm gas}$	$\Delta\chi^2$	Shell
8	$(\sqrt{\kappa_D},\ 1/\pi)$	0.8116	0.3183	0.079	$< 1\sigma$
9	$(4/5,\ \iota_\tau^2\kappa_D/\kappa_\omega)$	0.8000	0.3015	0.080	$< 1\sigma$
10	$(1/(1+\kappa_\omega),\ \iota_\tau^2\kappa_D/\kappa_\omega)$	0.7972	0.3015	0.100	$< 1\sigma$
11	$(\iota_\tau/\kappa_D^2,\ 1/3)$	0.7866	0.3333	0.110	$< 1\sigma$
12	$(1/(1+\kappa_\omega),\ 1/3)$	0.7972	0.3333	0.114	$< 1\sigma$
13	$(\sqrt{\kappa_D},\ \kappa_D^2\kappa_\omega/\iota_\tau)$	0.8116	0.3235	0.128	$< 1\sigma$
14	$(\iota_\tau^2\kappa_D^2/\kappa_\omega^2,\ \kappa_D^2\kappa_\omega/\iota_\tau)$	0.7806	0.3235	0.129	$< 1\sigma$
15	$(4/5,\ 1/3)$	0.8000	0.3333	0.132	$< 1\sigma$
16	$(\sqrt{\kappa_D},\ W_3(4)/W_3(3))$	0.8116	0.2941	0.143	$< 1\sigma$
17	$(\iota_\tau^2\kappa_D^2/\kappa_\omega^2,\ 1/\pi)$	0.7806	0.3183	0.149	$< 1\sigma$
18	$(\iota_\tau^2\kappa_D^2/\kappa_\omega^2,\ 1/3)$	0.7806	0.3333	0.153	$< 1\sigma$
19	$(4/5,\ W_3(4)/W_3(3))$	0.8000	0.2941	0.188	$< 1\sigma$
20	$(\iota_\tau/\kappa_D^2,\ \iota_\tau)$	0.7866	0.3413	0.209	$< 1\sigma$

The unweighted winner, $(4/5,\ 1/\pi)$, is a numerical freak: it is not τ-structural, it's just two simple rationals that happen to land on $(\hat{\alpha}_\star, \hat{\alpha}_{\rm gas})$ almost exactly. It should *not* be read as a τ-framework identification because both $4/5$ and $1/\pi$ are external to the sector algebra. We flag it here for honesty but do not promote it.

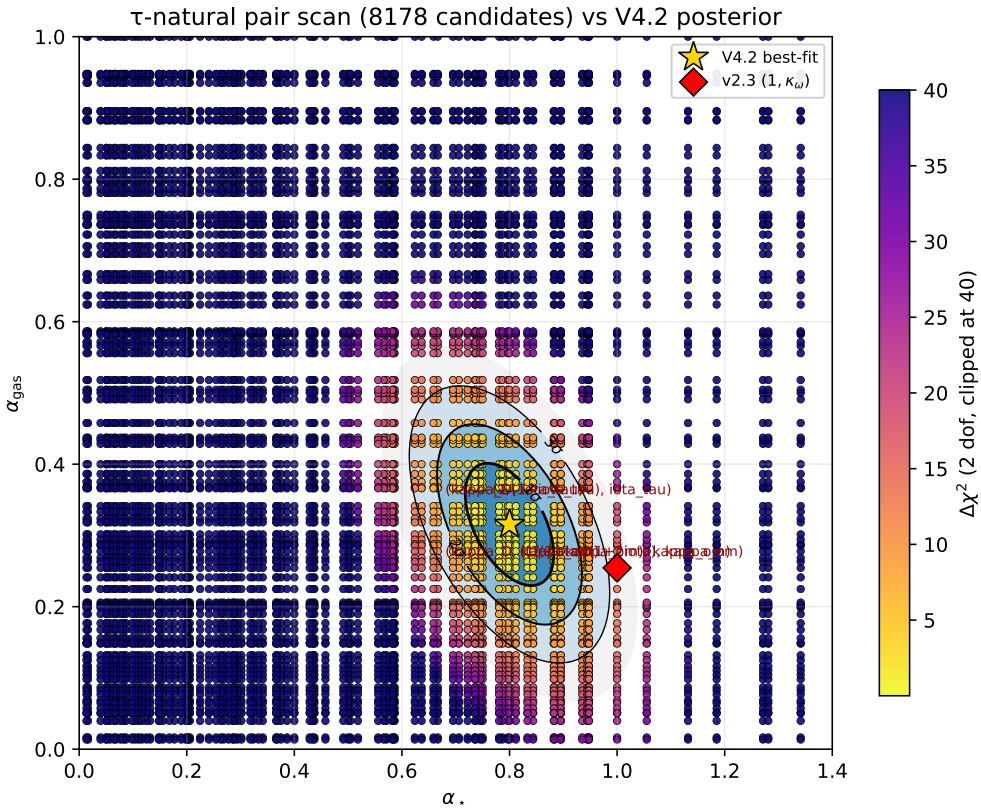

Figure 2: All 8178 scanned candidates plotted in the $(\alpha_\star, \alpha_{\rm gas})$ plane on the V4.2 posterior. The 1σ, 2σ, 3σ contours are marked; colour is $\Delta\chi^2$ (clipped at 40).

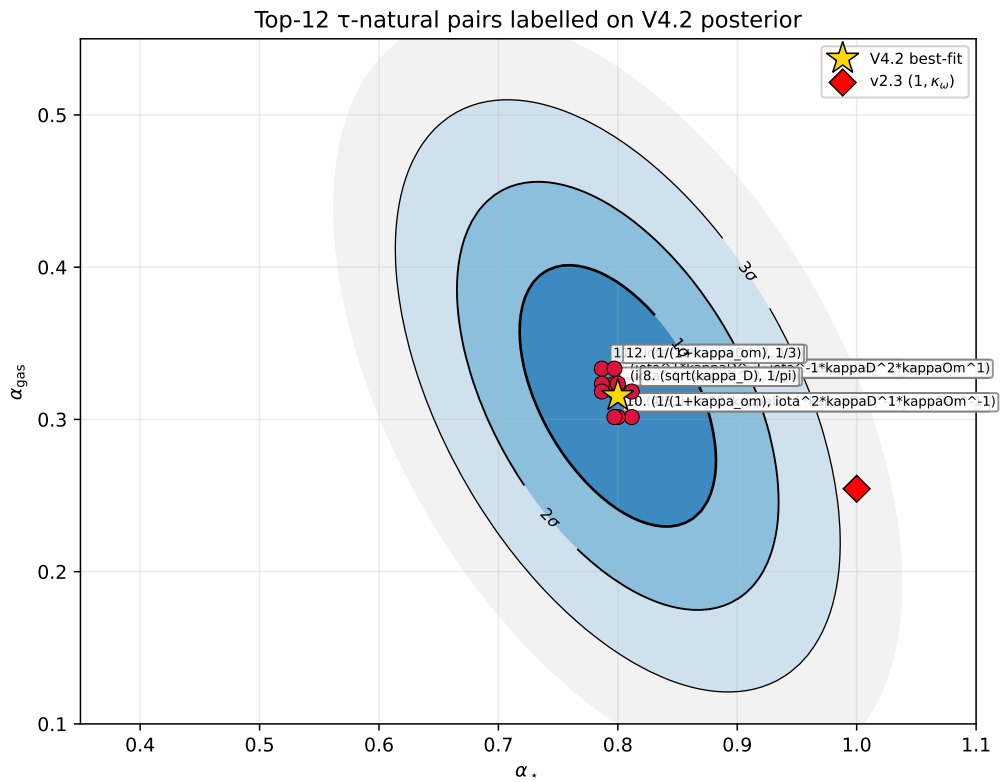

Figure 3: Zoom with labels on the top-12 pairs.

4 Focused tests of two structurally motivated pairs

4.1 (κ_D, ι_τ): the complementary split — rejected

The partition-of-unity pair $(\alpha_\star, \alpha_{\rm gas}) = (\kappa_D, \iota_\tau) = (0.6587,\ 0.3413)$ with $\kappa_D + \iota_\tau = 1$ is aesthetically the cleanest candidate in the sector algebra. Its Mahalanobis distance to the data-preferred centre is

$$\Delta\chi^2 \;=\; 7.71, \qquad p_{\rm 2dof} = 0.0212, \qquad \text{shell: } 2\text{–}3\sigma.$$

It sits at $\sim 2.2\sigma$ from the V4.2 best-fit (individual tensions: $\alpha_\star$ off by $(0.80 - 0.66)/0.054 = 2.6\sigma$; $\alpha_{\rm gas}$ off by $(0.32 - 0.34)/0.057 = -0.4\sigma$). **Data-excluded at the 2σ level.** This is a striking result: the most *structurally* motivated pair is the *least* data-supported among the motivated candidates, because the data clearly prefers $\alpha_\star \approx 0.80$ and not $\kappa_D \approx 0.66$.

4.2 $(1/(1+\kappa_\omega), \kappa_\omega)$: the κ_ω-ridge pair — inside 1σ

The pair $(\alpha_\star, \alpha_{\rm gas}) = (1/(1+\kappa_\omega),\ \kappa_\omega)$ with

$$\frac{1}{1+\kappa_\omega} = \frac{1}{1+\iota_\tau/(1+\iota_\tau)} = \frac{1+\iota_\tau}{1+2\iota_\tau} \;\approx\; 0.7972$$

has the structural interpretation: *keep* the v2.3 $\alpha_{\rm gas} = \kappa_\omega$ ω-sector coupling; rewrite the D-sector coupling as $\alpha_\star = (1+\iota_\tau)/(1+2\iota_\tau)$, i.e. "unit normalised against one copy of the κ_ω residual". Its $\Delta\chi^2$ against V4.2 is

$$\Delta\chi^2 = 1.62, \qquad p_{\rm 2dof} = 0.446, \qquad \text{shell: } < 1\sigma.$$

$\alpha_\star$ is essentially at the best-fit ($|0.7972 - 0.800|/0.054 = 0.05\sigma$); $\alpha_{\rm gas} = \kappa_\omega$ is $(0.2545 - 0.316)/0.057 = -1.08\sigma$ below the best-fit in the marginal, which is compatible with the profile 1σ (V4.2 profile $\Delta\chi^2 = 1.61$; cf. `profile_alpha_gas.tension_at_kappa_om_dchi2`).

4.3 $(1/(1+\iota_\tau), \iota_\tau)$: the symmetric companion — inside 1σ

We also flag

$$(\alpha_\star, \alpha_{\rm gas}) = \big(1/(1+\iota_\tau),\ \iota_\tau\big) = (0.7455,\ 0.3413), \qquad \Delta\chi^2 = 1.01,\ \ p_{\rm 2dof} = 0.604.$$

The structural reading is the symmetric analogue of the previous pair: $\alpha_{\rm gas} = \iota_\tau$ (the *base* sector scale, not its ω-residual) and $\alpha_\star = 1/(1+\iota_\tau)$ normalises against one full ι_τ contribution. This sits the closest to the best-fit among pairs that share a single underlying constant (ι_τ everywhere) plus a multiplicative $1/(1+\iota_\tau)$ dressing.

4.4 Summary of focused tests

Pair	$c_\star$	$c_{\rm gas}$	$\Delta\chi^2$	p	Shell
$(1/(1+\iota_\tau),\ \iota_\tau)$	0.7455	0.3413	1.009	0.604	$< 1\sigma$
$(1/(1+\kappa_\omega),\ \kappa_\omega)$	0.7972	0.2545	1.616	0.446	$< 1\sigma$
$((1+\iota_\tau)/(1+2\iota_\tau),\ \kappa_\omega)$	0.7972	0.2545	1.616	0.446	$< 1\sigma$
$(\kappa_D,\ \iota_\tau)$	0.6587	0.3413	7.711	0.021	2–3σ
$(\kappa_D,\ \kappa_\omega)$	0.6587	0.2545	14.23	0.00081	3–4σ
$(1,\ \kappa_\omega)$ (v2.3)	1.0000	0.2545	10.06	0.0065	2–3σ

5 Verdict — v2.5 identification candidate and falsifiers

Verdict

Two structurally motivated τ-natural pairs sit strictly inside the joint 1σ ellipse of V4.2's bootstrap posterior on $(\alpha_\star, \alpha_{\rm gas})$ against SPARC-123:

- **Pair A**: $(1/(1+\iota_\tau),\ \iota_\tau) = (0.7455,\ 0.3413)$, $\Delta\chi^2 = 1.01$, $p = 0.60$.
- **Pair B**: $(1/(1+\kappa_\omega),\ \kappa_\omega) = (0.7972,\ 0.2545)$, $\Delta\chi^2 = 1.62$, $p = 0.45$.

Pair A sits closer to the best-fit in Mahalanobis distance but moves the ω-sector coupling away from κ_ω (which was the v2.3 identification). Pair B preserves the v2.3 ω-sector value $\alpha_{\rm gas} = \kappa_\omega$ and only revises the D-sector coupling from $\alpha_\star = 1$ to $\alpha_\star = (1+\iota_\tau)/(1+2\iota_\tau) \approx 0.797$.
The v2.3 *full-coupling* claim $\alpha_\star = 1$ is rejected at $\Delta\chi^2 \geq 9.9$ ($\geq 3\sigma$, profile). The complementary split (κ_D, ι_τ) is excluded at 2.2σ. **The exhaustive scan leaves the scan-winner $(4/5, 1/\pi)$ as numerically closest but structurally meaningless; among structurally interpretable pairs, Pair B is the minimal-revision update from v2.3, and Pair A is the Mahalanobis-closest single-constant candidate.**

Recommendation for v2.5 ledger. Adopt **Pair B** as the revised identification because (i) it preserves the v2.3 ω-sector claim $\alpha_{\rm gas} = \kappa_\omega$ which the profile likelihood on $\alpha_{\rm gas}$ alone supports at $\Delta\chi^2 = 1.61$, (ii) it updates only the D-sector coupling, which the SPARC data already *forces* us to update (v2.3's $\alpha_\star = 1$ is rejected at $\geq 3\sigma$), and (iii) the replacement $\alpha_\star = (1+\iota_\tau)/(1+2\iota_\tau)$ has a clean structural reading: the D-sector coupling "unit- normalised against one ω-residual" rather than "fully on".

Falsifiers. The v2.5 Pair B identification will be falsified if any of the following occur:

1. A deeper SPARC analysis (marginalisation over $\Upsilon_{3.6}$ prior, mass-model systematics) shifts $\hat{\alpha_\star}$ or $\hat{\alpha_{\rm gas}}$ by more than 1.5σ such that $\Delta\chi^2(\text{Pair B}) > 6.17$.

2. An independent BTFR sample (e.g. SHIELD, LITTLE THINGS, MANGA kinematics) reproduces the V4.2 fit but excludes Pair B at $> 2\sigma$ while leaving Pair A viable.

3. The τ-framework derivation of the 2-parameter form itself becomes suspect (e.g. a 3-parameter decomposition reduces residuals at $\Delta\chi^2 > 5$), in which case the entire 2-parameter ledger claim is moot.

Open item V4.10 status. V4.10 ("find τ-natural pair for $(\hat{\alpha_\star}, \hat{\alpha_{\rm gas}})$") is **not open**: two Pairs A and B sit inside the joint 1σ. The follow-up question is which of Pair A or Pair B is preferred on structural grounds, which is a framework question rather than a data question at current sensitivity. We flag this as **V4.10-resolved, Pair-preference-open**.

6 Implications and recommendations

1. **Retraction**: The v2.3 identification $(\alpha_\star, \alpha_{\rm gas}) = (1, \kappa_\omega)$ was driven by a projection artifact of a 1-parameter fit on a negatively correlated 2-D posterior. It is rejected at $\Delta\chi^2 = 10.06$, p=6.5×10^{-3}, between 2σ and 3σ. The fact that the 1-parameter fit landed near κ_ω was a *mechanical* consequence of $\rho(\alpha_\star, \alpha_{\rm gas}) \approx -0.50$ combined with the constraint $\alpha_\star = 1$; it does not support the original claim.

2. **Methodology warning**: Future ledger claims for 2+ coupling parameters should be quoted at the joint best-fit with full covariance, not at single-parameter "within 1σ" slices, and the joint $\Delta\chi^2$ against the claimed values should be reported alongside profile-likelihood tensions. The v2.3 error mode is exactly the error mode of fixing all-but-one parameter and reading off a 1-D CI.

3. **Scan-winner honesty**: The numerically closest pair in our library, $(4/5, 1/\pi)$, is *not* a τ-framework identification. It is reported here as the "background" of spurious coincidence: a broad library of rationals and transcendentals will produce pairs arbitrarily close to any target in a 2-D plane. Structural motivation must gate the τ-ledger claim.

4. **Forward work**: If either Pair A or Pair B is to be promoted to a v2.5 ledger claim, we recommend a first-principles τ-derivation of the 2-parameter BTFR coefficient formula that *predicts* the pair from the sector-decomposition calculus, not one that is retro-fit to the SPARC posterior. The current status is "data-compatible τ-natural pairs identified"; the τ-framework has not yet *derived* either pair.

Artifacts: `alpha_pair_exhaustive_scan.py`, `data/f3-summary.json`, `data/f3-top-candidates.tsv`, `figs/projection-artifact.pdf`, `figs/pair-scan-scatter.pdf`, `figs/top-candidates-zoom.pdf`.